DAA EXPERIMENT 10

Closest Pair of Points – Brute Force

Aim

To find the closest pair of points in a two-dimensional plane using the Brute Force method.

Algorithm

1. Store the coordinates of all points.
2. Initialize the minimum distance to infinity.
3. Compare every pair of distinct points.
4. Calculate the Euclidean distance between each pair.
5. If the distance is smaller than the current minimum, update the minimum distance and closest pair.
6. Display the closest pair and minimum distance.
7. Stop.

Program

```
# Closest Pair of Points - Brute Force

import math

points = [(1, 2), (4, 6), (5, 3), (2, 3)]

min_dist = float("inf")
closest_pair = None

for i in range(len(points)):
    for j in range(i + 1, len(points)):
        x1, y1 = points[i]
        x2, y2 = points[j]

        dist = math.sqrt((x2 - x1)**2 + (y2 - y1)**2)

        if dist < min_dist:
            min_dist = dist
            closest_pair = (points[i], points[j])

print("Closest Pair:", closest_pair)
print("Minimum Distance:", round(min_dist, 2))
```

Complexity Analysis

- Best Case: $O(n^2)$
- Average Case: $O(n^2)$
- Worst Case: $O(n^2)$
- Auxiliary Space Complexity: $O(1)$

Output

```
Closest Pair: ((1, 2), (2, 3))
Minimum Distance: 1.41
PS C:\Users\jithe\OneDrive\Documents\DAA - programs>
```

Result

Thus, the closest pair of points was successfully found using the Brute Force method.